KAVACH — On-device scam shield for Indian phone users

Bharat AI-SoC Challenge 2026–27 · Problem Statement 3 (Cybersecurity & Digital Safety) · Phase 1 technical report

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1. Summary
2. Problem and scope
3. Architecture
   - 3.1 Pipeline A — speech (Guardian, Arm CPU; NPU in Phase 2)
   - 3.2 Pipeline B — reasoning (Llama 3.2 1B, ExecuTorch, Arm CPU)
   - 3.3 The always-on fast path (both phones)
   - 3.4 User's phone modules
   - 3.5 Calls on one phone: call listening, call awareness, KAVACH Phone
   - 3.6 Network Watch (third pipeline; details in `docs/network_watch.md`)
   - 3.7 OEM / system mode (`docs/oem_mode.md`)
4. Arm-specific engineering
5. Data
6. Engineering quality
7. Results
   - 7.1 Text classifiers (held-out test splits)
   - 7.2 Call Guardian (call level, replayed in call time)
   - 7.3 Noisy rooms (host, `bench/e2e_noise.py`)
   - 7.3b End-to-end audio (host, earlier run)
   - 7.4 Llama pipeline verification
   - 7.5 On the team's phone and through the real Hindi speech model
   - 7.6 Network Watch
   - 7.7 PS3 scope coverage
   - 7.8 Still to measure (scripts ready)
8. Limitations and threat model (summary)
   - 7.9 Llama 3.2 1B after LoRA fine-tuning (Colab, 12 Nov 2026)
9. Phase 2 plan
10. Reproducing everything

Team: Siddhant Singh (call listening: on-device speech recognition, Call Guardian and the Android app) · Bogavalli Sai Akhilash (detection and Llama: scam playbook, data, classifiers, LoRA fine-tuning and evaluation) · Shivansh Bose (Shield and Network Watch: messages, links, apps, payments, network and Wi-Fi checks, privacy and security review) · **Faculty mentor:** Dr. Rahul Ratnakumar. All three members worked across every part of the project; the roles name each person's main area.

1. Summary

KAVACH is **one app on one Android phone**. It listens to a suspicious call the user has put on speaker (with consent), understands it on the phone, and stops the victim before money moves. It joins what it hears with the SMS, link, app, screen-share, payment and network activity that follow, because real Indian scams move across channels. **Nothing leaves the device:** the app does not request the Android `INTERNET` permission.

KAVACH runs **three on-device pipelines** that feed one kill chain: **call listening** (the conversation), **Shield** (messages, links, apps, screen sharing, payments, and *that* a call with an unknown number is going on), and **Network Watch** (what installed apps send over the network, the website names they look up, and risky devices on the home Wi-Fi). Each one works alone; together they cover the whole scam.

What exists at the Phase 1 freeze (all in this repository):

- **One Android app + one optional app** (Kotlin, no AndroidX, built on every push by GitHub Actions and checked on an Android 14 emulator with screenshots of every screen): *KAVACH* (sentence-by-sentence Hindi/Hinglish and English ASR with Auto language, Call Guardian, Llama 3.2 1B advisor, live placement

check, **one-phone call listening** through Android's accessibility capture rule (§3.5), "Protect my calls" auto-start, Message Shield, Link Check, call awareness, app-traffic detector, App Inspector, Screen-share Guard, opt-in KAVACH Phone dialer, kill-chain hub, STOP card, on-phone performance benchmark), and the optional *KAVACH Network Watch* app (DNS watch and a home Wi-Fi check, §3.6). It is separate only because it needs the internet permission.

- **Measured on the team's phone (Xiaomi, Android 14):** a live English scam call played through a speaker reached STOP, and Llama 3.2 1B (SpinQuant INT4, ExecuTorch) explained it on the phone: first token 7.2 s with the warm system prefix, 10.1 tokens/s decode (§7.5).
- **A Python reference implementation** of every on-device algorithm, with the Kotlin port proven identical by **over 330 parity checks**, including tick-by-tick replays of whole calls.
- **The Llama pipeline** (LoRA data, fine-tuning, Hugging Face → Meta conversion, ExecuTorch 4-bit export ladder, evaluation), **verified end to end** on a tiny Llama-3.2-shaped model: conversion is exact (max logit difference 3.6e-7), and the exported `.pte` reproduces Hugging Face token by token through the KV cache.
- **Evaluation harnesses** and first results, all on held-out data.

Headline results. These cover the always-on path (rules + classifier) *without* Llama; no false STOP was raised on any genuine call in any set:

Evaluation set	Scam calls reaching STOP	STOP before the money/OTP request	Genuine calls wrongly STOPped
Synthetic test, 1,000 calls, phrasings never trained on	**90.6 %**	78.3 %	**0 %**
Same, caller heard through the speaker (3× ASR errors on the caller's lines)	71.1 %	55.6 %	**0 %**
Hand-written calls, 16 scam / 14 genuine	**93.8 %**	68.8 %	**0 %**
Locked held-out calls, 10 / 10, never used for tuning	**70 %**	50 %	**0 %**
Noisy-room audio: TTS call → speaker-phone → fan / TV / babble at 30–3 dB → ASR, 24 scam / 24 genuine	18 of 24 (6/6 at 30 and 15 dB)	—	**0 of 24**

Network Watch (§7.6): all 5 planted threats found in a replayed capture with no alert on normal browsing; in 200 simulated traces per scenario, spyware uploads and remote-control heartbeats are caught every time, with 0 % false alarms on normal app use.

The honest reading: the fast path is safe, with no false STOPs anywhere, but on unseen phrasing it leaves about 10 % of scam calls at CAUTION (about 30 % when the caller is badly heard). That gap is what the Llama 3.2 1B layer is for: it judges every CAUTION and may escalate it to STOP. Measuring that on phones is the first Phase 2 milestone (§9). The other bottleneck is speech recognition when a TV or other people are talking in the room (§7.3).

2. Problem and scope

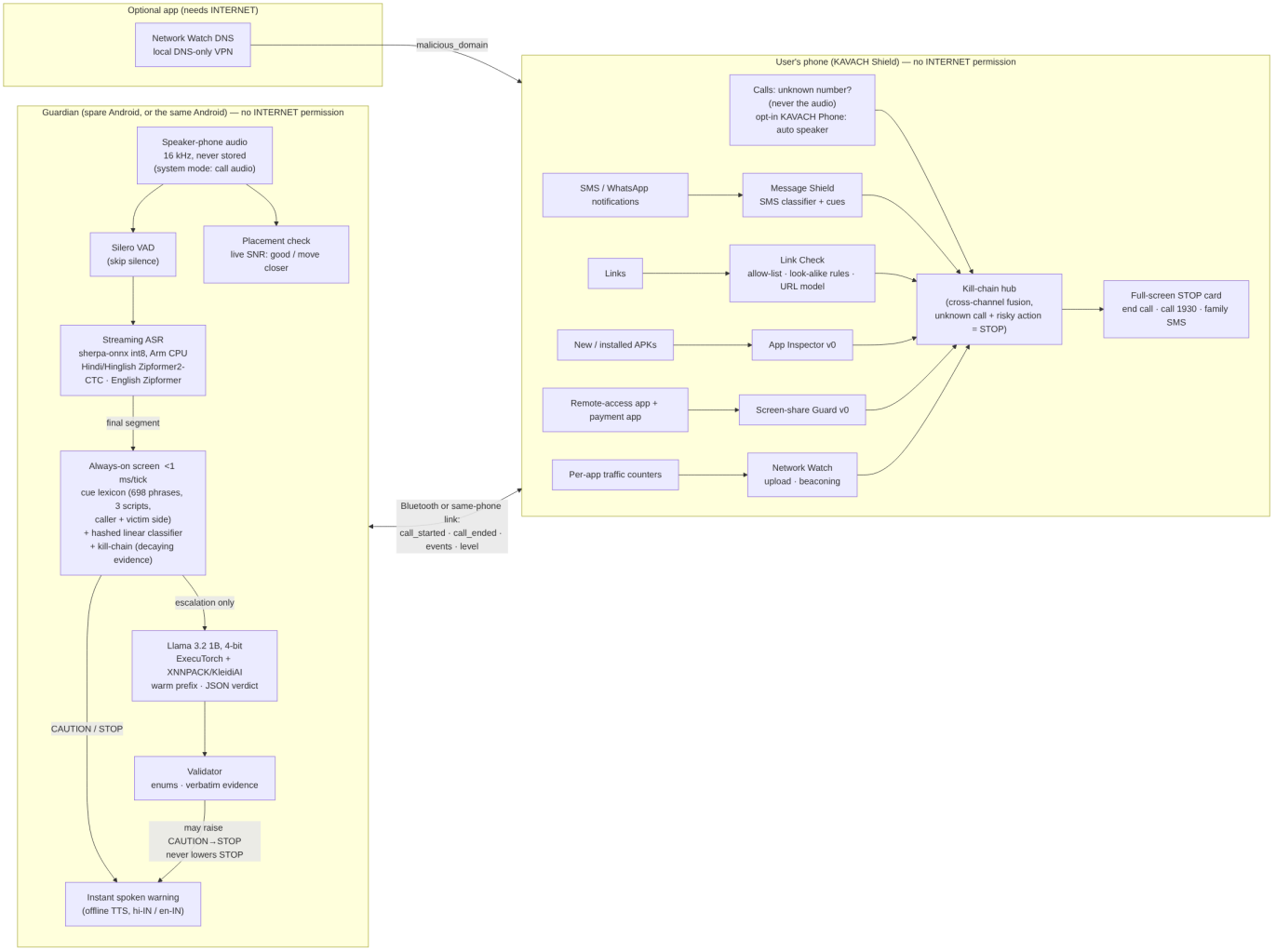
Indians lost ₹22,495 crore to cyber fraud in 2025 (MHA/I4C). "Digital arrest" calls alone caused 1.24 lakh cases and ₹1,918 crore of losses in 2024. A scam is a *sequence*: an authority claim, a threat or lure, isolation ("don't disconnect, don't tell your family"), an action (OTP, link, APK, screen share) and a payment. Single-

channel filters see one fragment. On-device call-scam detection exists only on premium phones. KAVACH targets the budget and mid-range Android phones most families own, explains every alert in Hindi or English, and gives the next step (hang up, call 1930).

Since Android 10, third-party apps cannot capture call audio. The **Guardian** therefore listens to the call on speaker, from a spare Android or from the same Android when the call is on another phone. That has three costs, and Phase 1 now addresses each of them:

Drawback	What KAVACH does about it	Section
Needs an extra device	**One-phone mode**: Shield alone knows *that* an unknown-number call is going on and stops risky actions during it; **OEM/system mode**: the same Guardian APK reads the call audio directly when a phone maker ships it as a privileged app	3.5, 3.7
The victim must put the call on speaker	**KAVACH Phone** (opt-in dialer role) puts unknown callers on speaker itself after a spoken notice; otherwise a “turn on speaker” nudge when a Guardian is linked	3.5
Noisy rooms	a bigger English recogniser (3× more noisy scam calls reach STOP), a live **placement check**, victim-side cues (the victim sits next to the Guardian), and fusion with actions that need no audio at all	3.1, 7.3

3. Architecture



System architecture: Guardian, Shield and Network Watch feed one kill chain; no network in the main apps

3.1 Pipeline A — speech (Guardian, Arm CPU; NPU in Phase 2)

Stage	Implementation	Notes
Capture	`AudioRecord`, `VOICE_RECOGNITION` source, 16 kHz mono, 100 ms chunks	AGC/noise suppression off; audio never written to storage
VAD	Silero VAD (sherpa-onnx), 512-sample windows	decoding is skipped after 1.5 s of silence; audio is still buffered, so nothing is lost
Streaming ASR — Hindi/Hinglish	Zipformer2-CTC, int8, Devanagari characters (`betterflow-hindi-streaming-ctc`, Apache-2.0)	writes English loan words in Devanagari; the cue lexicon covers both scripts
Streaming ASR — English	Zipformer transducer `en-2023-06-21`, int8, 188 MB (k2-fsa); the 20M model remains for low-end phones (`push_models.sh --asr-small`)	host RTF 0.12 (20M: 0.05); chosen by the noisy-room benchmark, §7.3
Placement check	20 ms frame powers over the last 5 s; 90th vs 10th percentile → SNR estimate; “good / move closer / too noisy” on screen	no audio kept; thresholds calibrated in §7.3
Endpointing	trailing silence 0.8 s, or 12 s maximum utterance	each final segment triggers an immediate engine tick

Design decision from this phase. The proposal named a Hindi FastConformer streaming model. Its maintainers document that it needs a patched sherpa-onnx feature extractor and cannot write Latin script (12 WER points worse on Hinglish). We therefore ship the Zipformer2-CTC model, which runs on the stock runtime and handles Hinglish. The NPU export of speech models (ExecuTorch Qualcomm/MediaTek backends) is Phase 2.

3.2 Pipeline B — reasoning (Llama 3.2 1B, ExecuTorch, Arm CPU)

Item	Choice
Model	Llama-3.2-1B-Instruct + KAVACH LoRA (r = 16, all projections), merged before export
Runtime	ExecuTorch `LlmModule` (Android AAR), XNNPACK delegate; KleidiAI int4 micro-kernels are enabled by default on Arm
Quantisation (shipped)	8da4w: 4-bit groupwise weights (g = 32), 8-bit dynamic activations, 4-bit embeddings
When it runs	only on escalation to CAUTION/STOP; never in the per-segment loop
Latency tricks	**warm prefix** (constant system prompt pre-filled into the KV cache at start and re-armed in the background after every alert); **early stop** as soon as the JSON object closes; greedy decoding; 8 s hard timeout
Thread	dedicated `kavach-llm` thread at urgent priority, so the scheduler places bursts on big cores; placement is logged from `/proc/self/task/*/stat`
Contract	ONE JSON object: `scam, type, stage, confidence, evidence[≤3 verbatim quotes], say(≤35 words), action`
Validator	enums enforced, confidence clamped, **quotes not found verbatim in the transcript are dropped** (and confidence capped), one re-prompt on invalid output
Authority	may raise CAUTION → STOP ("llm-early-warning") at confidence ≥ 0.8 and stage ≥ isolation; can never lower a rule STOP

3.3 The always-on fast path (both phones)

1. **Cue lexicon** (`assets/cues.json`): 698 phrases in 16 categories (authority, identity misuse, threat, urgency, isolation, credential request, remote access, link, payment, reward, family emergency, safety advice, and three **victim-side** categories: the person next to the Guardian reading out an OTP or account number, saying they are sending money, or begging not to be arrested), each in English, romanised Hinglish and Devanagari. The phrase rules support prefixes and gaps of 0–4 words. A request is ignored when a nearby word negates it ("never share your OTP" is *safety advice*, not a request).
2. **Hashed linear classifiers** (`cue_model`, `sms_model`, `url_model`). Features: FNV-1a hashed words, word pairs and 3–5 letter fragments, plus cue-category hits and category pairs. The models are 1 MB each, with identical inference in Python and Kotlin. Every score breaks down into per-word contributions, which become the "evidence" words shown to the user.
3. **Kill-chain** (`killchain.py` / `KillChain.kt`): evidence fades with a half-life that depends on the channel (10 min for calls, 30 min for SMS and links, 24 h for apps). The level (none → info → caution → stop) is *sticky* within a session: a scammer cannot calm the guard with small talk. STOP rules: isolation + authority; authority + identity + threat; two different action/payment requests; a hook plus an action with no genuine safety advice heard; **victim complying** (the victim reads an OTP or says they are paying *and* a scam pretext is present: police, identity misuse, threat, bank, or the victim's own fear; a delivery OTP or money for family alone never stops); **call + action** (a suspicious conversation and a risky action in the same call); **unknown call + risky action** (one-phone mode); cross-channel evidence; total score. CAUTION also for a payment app opened during an unknown-number call longer than 3 minutes. Every rule has a plain Hindi/English reason shown on both screens ("Unknown caller + a dangerous link").

4. **Guardian logic**: ticks after every final ASR segment and at least every 3 s. The classifier adds CAUTION or STOP when it stays above 0.8 for two consecutive ticks. It can also **veto** a weak rule-only STOP while it has been confidently benign (< 0.5); strong rules are never vetoed. Thresholds are tuned on the development sets only (`bench/results/guardian_tuning.json`).

3.4 User's phone modules

- **Message Shield** reads SMS/WhatsApp/Telegram *notifications*, so it needs no READ_SMS permission. It runs the SMS classifier, the cues and Link Check on every URL.
- **Link Check** checks, in order: an offline allow-list of genuine Indian domains, including the RBI-mandated `.bank.in` and `.fin.in` suffixes; then look-alike rules (brand name in an unofficial domain, typosquat by edit distance, homoglyph or punycode, shortener, raw IP, `@` trick, APK download, scam-prone TLD, brand page on a free website builder); then the URL model with a threshold chosen on dev for ≤ 3 % false alarms.
- **Kill-chain hub** fuses messages, links, apps, screen sharing and the Guardian's call level. It raises a full-screen STOP card (spoken, with a one-tap call to 1930) and can optionally SMS a family member.
- **App Inspector v0** (preview) scores the permission profile of each installed or new app: SMS + Accessibility + overlay, sideloaded, named like a bank/KYC/challan app. **Screen-share Guard v0** (preview) raises STOP when a payment or banking app comes to the foreground while a remote-access app is active.
- **Device link**: Bluetooth RFCOMM carrying newline-delimited JSON events when the Guardian is another phone. A **same-phone link** (explicit broadcasts protected by a signature-level permission) is used when Guardian and Shield are installed on one Android phone and the call is on another phone, e.g. an iPhone, on speaker. This is our team's demo setup. Both work offline.
- **Own-voice filter**: the Guardian ignores ASR segments recognised while it is speaking a warning and for 1.2 s afterwards. Otherwise its own warning ("police", "arrest", "1930") could raise its level.

3.5 Calls on one phone: call listening, call awareness, KAVACH Phone

- **Call listening on the same phone.** During a phone call Android gives the microphone to the call and every other app records silence; the one documented exception is an accessibility service (Android "Sharing audio input": *the call always receives audio; the app can capture audio if it is an accessibility service*). KAVACH ships `CallListenAccessService`, which requests no screen content and performs no actions; while it is enabled, the KAVACH app may keep hearing a call the user has put on speaker. With **Protect my calls** on, the microphone foreground service is started while a KAVACH screen is visible (Android's rule for later background use) and listening begins when an unknown-number call starts. During a real call the warning is not spoken (it would go into the call and mute KAVACH's own listening): the phone vibrates and the full-screen STOP card appears. Status: verified on an emulator for the UI and state flow; live-call capture on one phone is verified on real calls (§7.5).
- **Call awareness** (`CallMonitor`, phone-state broadcast): Shield knows a call is on, whether the number is in the contacts, and how long it has lasted. It never hears the call.
- **One-phone mode:** during an unknown-number call, a dangerous link, a risky APK, screen sharing, a suspicious upload or a malicious domain is STOP even with no Guardian.
- **Inside the one app**, the Shield part and the call listener exchange the same JSON messages they would send between two phones, in memory (`InProcessLink`); a suspicious conversation plus a risky action in the same call is STOP. A Bluetooth link to a second phone remains available for a family member's phone. Llama is loaded once and shared by calls, messages and network explanations (two copies would not fit in a phone's memory).
- **KAVACH Phone** (opt-in dialer role, `InCallService`): big-button call screen with the live KAVACH level; for an unknown number it says "KAVACH is turning on the speaker so the Guardian can listen" and routes

the call to the speaker; on STOP a one-tap **End call**.

3.6 Network Watch (third pipeline; details in `docs/network_watch.md`)

- **App traffic watch** inside Shield, **no INTERNET permission**: `NetworkStatsManager` per-app byte counters every 30 s (also with the screen off). Only *suspect* apps (sideloaded, new, remote-access, or rated suspicious) are judged, so normal apps stay quiet; "I know this app" trusts one. Kinds: `suspicious_upload` (≥ 5 MB in 5 min in the background, or a slow ≥ 4 MB/hour trickle) and `beaconing_app` (small background bursts at a steady rhythm).
- **Optional DNS-watch app** (`io.kavach.netwatch`): a local DNS-only `VpnService` learns which app looks up which name, forwards the lookup unchanged to the phone's own DNS server, and flags bank look-alikes and bursts of machine-generated names that do not exist (malware looking for its control server) as `malicious_domain`. It must hold INTERNET to forward lookups, so it is a separate opt-in app.
- **Flow model** (1,665-parameter MLP, ExecuTorch `.pte` 11 KB) trained on UNSW-NB15. It is evaluated and reported but **raises no alerts**, because UNSW-NB15 has no normal HTTPS traffic (§7.6).
- Llama explains a network finding with the same JSON contract (`network_user_prompt`).

3.7 OEM / system mode (`docs/oem_mode.md`)

The Guardian manifest asks for `CAPTURE_AUDIO_OUTPUT`. A normal install never gets it. A phone maker that ships the Guardian as a privileged system app grants it; the Guardian then reads `VOICE_CALL` (both sides), falling back to `VOICE_DOWNLINK` and then the microphone, runs on the same phone as the call, and starts by itself on unknown calls. This removes both the spare phone and the speaker step, but needs an OEM partner, depends on the phone's audio HAL, and needs explicit user consent. We cannot demonstrate it on a retail phone.

4. Arm-specific engineering

Lever	Status in Phase 1	How it is measured
ExecuTorch + XNNPACK, `.pte` export	export ladder verified on a tiny Llama-3.2 model; Android integration compiled against `LlmModule`	`models/llama/tiny_pipeline_check.py`, `bench/results/llama_pipeline_check.json`
KleidiAI int4 kernels on vs off	build script for two `llama_main` binaries (`EXECUTORCH_XNNPACK_ENABLE_KLEIDI=ON/OFF`)	`bench/build_llama_runner.sh`, `bench/llama_bench.sh` (same model, prompt and phone)
Quantisation ladder	bf16 → 8da4w g128 + 8-bit embeddings → 8da4w g32 + 4-bit embeddings	`models/llama/export_ladder.sh` + `eval_llama.py` (quality) + `llama_bench.sh` (speed, memory)
big.LITTLE placement	`taskset` masks for big, little and all cores × 2 or 4 threads in the bench; the app logs the core each thread last ran on	`bench/llama_bench.sh`, Guardian session CSV
Battery and thermal	Guardian writes a perf row every 5 s: per-core clock, battery %, °C, current, thermal status, PSS, ASR RTF, Llama TTFT and tok/s	`bench/parse_session.py`, `bench/thermal_soak.sh` (30 min)
Prefix caching	warm system prefix in the KV cache via `prefillPrompt`; falls back automatically if a runtime build lacks text prefill	Llama stats in the session log (`usedWarmPrefix`)
Fast path on LITTLE cores	0.44 ms per tick (lexicon + kill-chain + classifier over 120 tokens) in the shipped Kotlin code on a laptop JVM	`bench/results/kotlin_engine_host.json`; on-device numbers from the perf overlay

One finding from this phase: `qmode: int8` fails to export on the GQA projections in ExecuTorch 1.5.1, and `torchao:8da8w` cannot be combined with XNNPACK. The ladder's middle rung is therefore 4-bit with coarse groups. Both failures were reproduced on the tiny model.

5. Data

Need	Source	Use
Call transcripts	**Synthetic generator** (`data/synth/generate_calls.py`): 11 scam scripts and 9 benign scenarios built from 47 utterance pools, including victim-side lines (reading an OTP, “main transfer kar raha hoon”, “mujhe arrest mat karo”) with benign twins (a delivery OTP, money for a relative). Each pool has an English, Hinglish and Devanagari rendering, with ASR-style corruption and realistic timing. Scripts are paraphrased from I4C, RBI and bank advisories.	train 8,000 / dev 600 / test 1,000 calls
Hard negatives	genuine bank fraud alerts that say “never share OTP”, delivery agents asking for the delivery OTP, relatives asking for money, people *talking about* scams, genuine police calls, office talk about payments and screen sharing	in every split
Held-out phrasing	**template-disjoint splits**: each pool is partitioned by item into train, dev and test, so dev and test contain phrasings never trained on	model selection on dev; test reported once
Caller heard through the speaker	`test_far`: the test calls with 3× ASR errors on the caller’s lines only (the victim sits next to the Guardian mic)	reported, never tuned
Independent checks	**30 hand-written calls** (a different author style) and **20 locked held-out calls**, written before any tuning against them and used only for reporting	`data/dev/`
Real ASR noise	88 English training calls synthesised with Piper TTS, passed through a speaker-phone channel, transcribed by the Guardian’s ASR, with stage labels aligned by time	`models/asr/make_asr_noisy_calls.py`
SMS	UCI SMS Spam Collection (CC BY 4.0) + 4,300 synthetic Indian scam and genuine messages (KYC, e-challan APK, electricity, parcel fee, UPI refund, jobs, loans vs real OTP and debit-alert formats)	`sms_model`
URLs	Phishing.Database active phishing domains (MIT, 40k sampled), top-10k popular domains, the genuine Indian domain list, 8k generated Indian look-alikes; split by registrable domain	`url_model`
Network	UNSW-NB15 (Moustafa & Slay 2015) and NSL-KDD (Tavallaee et al. 2009) for the flow model; 25 DGA families (Cucchiarelli et al. 2021, Netlab 360) +	`docs/network_watch.md`

Need	Source	Use
	Alexa for the domain model; simulated per-app traffic traces	
Llama fine-tuning	4,000 / 400 instruction pairs generated by the same prompt code the phone uses (`models/llama/make_sft_data.py`)	LoRA

No real victim audio or personal data is used or committed. The final test set must be **human-recorded** role-play calls (`demo/recording_protocol.md`).

6. Engineering quality

- **Single source of truth**: lexicon, playbook, domain lists and model weights live in `assets/` and are packaged into both APKs by Gradle.
- **Parity**: `tests/make_parity_vectors.py` writes golden outputs from Python, and `android/core` checks 336 of them. The checks cover normalisation, hashing, lexicon hits, feature vectors, logits of all three models, URL verdicts, message verdicts, kill-chain steps, retrieval, prompts, JSON validation, full Guardian replays, Network Watch traffic traces, domain and flow model scores, DNS parsing and placement SNR. Porting found and fixed two real bugs: Python's brand ordering depended on hash randomisation, and Java and Python IDNA decoders disagree on some punycode.
- **Unit tests**: 35 pytest tests. Writing them found a safety bug: `"scam": "maybe"` was being read as *not a scam*; it now fails validation and triggers the re-prompt.
- **Fail-safe by construction**: if the model is missing, slow or throttled, the rule + classifier path still speaks a warning, which is generated from the vetted playbook.
- **Privacy by construction**: no `INTERNET` permission in the main apps, audio never stored, transcript kept in memory only, notifications read without SMS permission, network watched through byte counters (no contents, no destinations) unless the user installs the optional DNS app.

7. Results

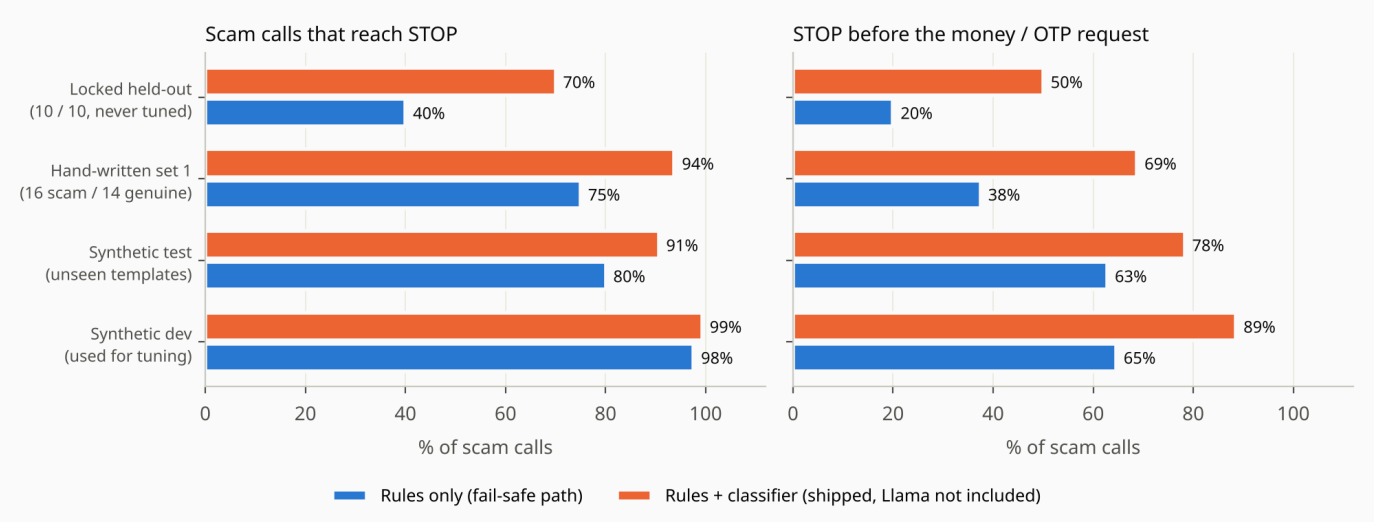
7.1 Text classifiers (held-out test splits)

Model	Task	Test n	Precision	Recall	F1	ROC-AUC
`cue_model`	120-token call window → scam	9,010	0.951	0.867	**0.907**	0.960
`sms_model`	SMS / notification → scam	1,914	0.996	0.888	**0.939**	0.990
`url_model` (rules + model)	URL → phishing	10,196	0.987	0.752	0.853	0.945

Model selection on a *template-disjoint* dev split mattered. With a dev split that shared templates with training, the cue model looked perfect on dev (F1 0.999) but reached only F1 0.817 on test. Adding cue-category features raised test F1 to 0.903, and the victim-side categories to 0.907. (Retraining with the new lexicon moved the SMS model from F1 0.943 to 0.939; we kept the retrained model so its features match the lexicon.)

Link Check flags 98.5 % of generated Indian look-alikes, **0 %** of genuine Indian domains, 3.9 % of popular global domains and 69 % of generic global phishing domains (many are random strings with no brand cue; at a ≤3 % false-alarm budget these stay “unknown”).

7.2 Call Guardian (call level, replayed in call time)



Call-level results

Set	Config	STOP recall	≥ CAUTION	STOP before money	Median time to STOP from first scam cue	False STOP	False CAUTION
Synthetic test (511 scam / 489 genuine)	rules only	80.0 %	89.2 %	62.8 %	17.6 s	0 %	0 %
	shipped	**90.6 %**	97.3 %	78.3 %	12.7 s	**0 %**	7.2 %
Test, caller heard through the speaker (502 / 498)	rules only	60.8 %	68.7 %	41.0 %	19.3 s	0 %	0 %
	shipped	**71.1 %**	88.8 %	55.6 %	14.7 s	**0 %**	4.6 %
Hand-written (16 / 14)	rules only	75.0 %	87.5 %	37.5 %	24.7 s	14.3 %	21.4 %
	shipped	**93.8 %**	93.8 %	68.8 %	17.0 s	**0 %**	21.4 %
Locked held-out (10 / 10)	rules only	40 %	70 %	20 %	21.9 s	0 %	10 %
	shipped	**70 %**	80 %	50 %	22.6 s	**0 %**	40 %

Effect of the victim-side cues (same regenerated data, before → after): synthetic test STOP 88.3 → 90.6 %, caller-through-speaker 68.5 → 71.1 %, hand-written false CAUTION 28.6 → 21.4 %, locked held-out unchanged (70 % / 0 % / 40 %; run once each, for reporting only), no new false STOP anywhere. The synthetic data was regenerated with the new victim lines, so these test numbers are not directly comparable with the earlier 86.7 %.

The proposal targets (median ≤ 30 s to STOP and ≤ 2 % false STOPs) are met on every set. The ≥ 90 % “caught before payment” target is **not met yet** (50–77 %), and false CAUTIONs on genuine calls are high on the small hand-written sets. Both are the job the Llama layer was designed for, and we will measure it in

Phase 2 rather than claim it now. The classifier veto removed both rule-only false STOPs on the hand-written set (people *discussing* a scam; an office call about screen sharing and payments).

7.3 Noisy rooms (host, `bench/e2e_noise.py`)

Four scripted calls (2 scam, 2 genuine) are synthesised with Piper TTS, passed through a telephone channel and room reverb, and mixed with **fan/traffic hum, a TV (another voice reading news, plus music) or several people talking**, at 30, 15, 8 and 3 dB: 48 noisy calls per configuration.

English recogniser + front-end	Scam STOP 30 / 15 / 8 / 3 dB	Genuine STOP	WER hum / TV / babble	Host RTF
20M (previous default)	3/6 · 3/6 · 0/6 · 0/6	0/24	0.53 / 0.80 / 0.74	0.05
20M + GTCRN speech denoiser	1/24 in total	0/24	0.48 / 0.97 / 1.13	+0.08
20M + scam-word hotwords	3/24 in total	0/24	0.60 / 0.74 / 0.75	
en-2023-06-21 (new default)	**6/6 · 6/6 · 4/6 · 2/6**	**0/24**	**0.20** / 0.83 / 0.90	0.12
en-2023-06-21 + hotwords	6/6 · 6/6 · 5/6 · 2/6	0/24	0.19 / 0.85 / 0.86	0.12

Decisions from the data: the bigger English model ships (3× more noisy scam calls reach STOP). The denoiser stays **off**: it helps on steady hum but invents speech-like fragments from TV and babble (WER above 1). Hotwords stay off (+1 call, not worth the beam-search cost; the Hindi model is CTC, which has no hotwords anyway). With a TV or people talking, even the bigger model transcribes the *background voices*: a "which voice is the call?" problem, which is why the Guardian shows a **placement check**. Its SNR estimate reads about 5 dB high (true 30/15/8/3 dB read as 35/21/14/10 dB), and its thresholds are set from these runs (≥ 18 dB is "good"). There is one TTS realisation per condition, so the numbers are small: read them as a direction, not as precise rates. (The 20M rows with denoiser / hotwords used the cue model from before the victim-side change; the two base rows use the current one.)

7.3b End-to-end audio (host, earlier run)

The scripted calls are synthesised with Piper TTS, passed through an 8 kHz telephone channel (300–3,400 Hz), room reverb and noise at 30, 15 and 8 dB SNR, then streamed into the Guardian's English ASR and engine:

Call	WER (30 / 15 / 8 dB)	Level reached
Digital arrest	0.38 / 0.49 / 0.61	STOP / STOP / CAUTION
Fake KYC	0.30 / 0.26 / 0.49	STOP / STOP / CAUTION
Genuine bank fraud alert	0.41 / 0.45 / 0.54	none / none / none
Delivery agent asking for OTP	0.27 / 0.32 / 0.59	none / info / none

ASR real-time factor on the host is 0.05–0.07. The small generic English model mishears Indian scam vocabulary ("money wandering" for *laundering*, "arbi" for *RBI*, "o p" for *OTP*). We also trained the cue model on real ASR transcripts of training calls and compared the two models on **identical saved audio**: one of the six scam runs improved (digital arrest at 15 dB, INFO → CAUTION), none got worse, and genuine calls were unchanged (`bench/results/e2e_audio_paired.json`). One fake-KYC run at 8 dB still only reaches INFO. The table above comes from a fresh audio run; Piper TTS is stochastic, so levels vary between runs. The real fix is better ASR (§8).

Negative result, kept out of the product. We tried “acoustic lexicon expansion”: synthesise each cue phrase on its own and add its ASR misrecognitions as aliases. On short isolated phrases the streaming model returned word fragments (“ation”, “ly”), which would cause false alarms, so the aliases were discarded.

7.4 Llama pipeline verification

Check	Result
HF → Meta weight conversion (tiny Llama 3.2-shaped)	max $\|\Delta\text{logit}\|$ 3.6×10^{-7}
fp32 `.pte` with KV cache and custom SDPA vs HF, step by step	max $\|\Delta\|$ 4.8×10^{-7}, 100 % next-token agreement
shipped 8da4w `.pte` vs HF	100 % next-token agreement
Real alert prompts	1,361-character constant prefix + 1.7–2.0 k-character suffix (`bench/prompts/`)

7.5 On the team’s phone and through the real Hindi speech model

Live tests on a Xiaomi phone (Android 14), synthetic voices played through a laptop speaker 10–20 cm from the phone:

Test	Result
English digital-arrest scam call	STOP; Llama explanation on the phone, 106 tokens, first token 7.2 s (warm prefix), 10.1 tok/s
Llama self-test (cold)	108 tokens, first token 15.6 s, 11.6 tok/s
English genuine bank call, delivery call	not red
English KYC and digital-arrest calls (first run)	yellow; cause found: the spoken yellow warning muted the microphone for 19–33 s and the decisive sentences were lost. Fixed (short spoken warnings; spelled-out letters “o t p” joined); both reach STOP when replayed through the same English model on the build machine
Hindi calls	inconsistent at first (a genuine call went red; a scam call stayed silent); investigated below

Caveats found and kept visible: Llama’s advice was at first identical to the worked example in its instructions; such wording is now replaced by the playbook text and labelled “advice from KAVACH’s playbook”, and a copied answer cannot raise a warning. LoRA fine-tuning (§7.9) takes copied advice from 32 % to 0 %.

Hindi through the speech model. 100 unseen synthetic Hindi calls (50 scam, 50 genuine) spoken by Piper TTS, passed through a channel and decoded by the same streaming Hindi model the phone uses:

Condition	Build	Scam → red	Scam → yellow or red	Genuine → red
near speaker, quiet room	before	24 %	58 %	0 %
near speaker, quiet room	after	44 %	88 %	0 %
telephone band, SNR 10–30 dB	before	0 %	12 %	0 %
telephone band, SNR 10–30 dB	after	0 %	44 %	0 %

A better Hindi speech model. AI4Bharat's IndicConformer Hindi (120 M parameters, CTC; MIT licence; int8 ONNX export for sherpa-onnx, 198 MB) recognises each sentence cut out by the VAD instead of streaming word by word. Same 100 calls, same audio:

Condition	Streaming Hindi model: red / yellow-or-red	IndicConformer: red / yellow-or-red	Genuine → red / yellow (IndicConformer)
near speaker, quiet room	44 % / 92 %	**66 % / 96 %**	0 % / 6 %
telephone band, SNR 10–30 dB	0 % / 46 %	**58 % / 86 %**	0 % / 6 %

The three genuine calls that reach yellow are a real traffic-police call and two real card-fraud checks from a bank, the closest benign twins of scams. The team's two Hindi test files: scam red 12/12, genuine never yellow or red. Host speed: 0.04 × real time on 2 threads (the phone figure comes from the benchmark screen). The app uses IndicConformer whenever its files are on the phone (`scripts/push_models.sh --asr-hi-big`) and falls back to the streaming model otherwise.

Earlier changes, streaming model: 208 mixed-script word-list variants (the model writes "bank", "otp", "card" in Latin letters inside Hindi), the classifier retrained with 410 Hindi calls as the model hears them, "tell no one" treated as safety advice when the sentence is about an OTP or PIN, and the `isolation+authority` rule made vetoable by a confidently benign classifier. Text-only sets are equal or better with 0 false STOPs. The team's two Hindi test files, replayed 12 times each in varied near-speaker conditions: scam red 12/12; genuine calm 10/12, yellow 2/12, red 0/12. **Reading:** English is far stronger than Hindi, and the reason is the small streaming Hindi speech model (69 MB vs 188 MB for English): with telephone-band noise its text is mostly unusable. Training and test audio share one TTS voice, so human voices will score lower.

Retest on the phone, 12 Nov 2026 (IndicConformer Hindi + Parakeet English installed). The team's six test files, played from a laptop speaker 10–30 cm from the phone: Hindi digital arrest **red** (twice), Hindi genuine bank call lowest hint only ("worth watching", caused by one misheard word), English digital arrest **red**, English KYC **red**, English delivery call and English genuine bank call **no warning**. 6/6 as expected. Yellow warnings are no longer spoken, so the microphone is never muted before the decision (the 19–33 s gap above is gone).

Real phone calls, one phone (12 Nov 2026, 03:15–03:53): 3/3 as expected. A teammate called the test phone from a number not in its contacts over the normal cellular network; the call was answered in the phone's own dialer and put on speaker; KAVACH started listening by itself (call language: Auto). The teammate read the role-play scripts (`demo/roleplay_scripts.md` style).

Call	Result
Hindi digital-arrest scam (fake police, Aadhaar parcel, arrest warrant, “tell no one”, RBI safe account)	**STOP** during the call; transcript through the phone line was close to the script; full-screen Stop card
English KYC scam (SBI KYC, account blocked, link by SMS, OTP)	**STOP** during the call; Auto chose English; scam type “Fake KYC / bank account block”
Genuine bank fraud check in Hindi (card transaction, card blocked, “never share your OTP or PIN”)	**no warning** (“no warning signs so far”) for the whole call

What went wrong first and was fixed: when the phone’s default dialer app was changed, Android stopped KAVACH (“permissions revoked”) and marked the call-listening accessibility service as crashed; the switch still showed “on”, but every recording during a call was silenced (seen in `dumpsys audio`: “silenced” exactly for each call). Turning the service off and on restored it. The app now detects both states (service switched on but not running; recording silenced during a call) and tells the user how to fix it.

Better English speech model and Auto language. NVIDIA Parakeet TDT-CTC 110M (CC-BY-4.0, int8, 126 MB) now recognises English sentence by sentence, like IndicConformer for Hindi. With the call language on **Auto** (the default) every sentence is recognised by both models and `kavach_core/langpick.py` keeps the transcript whose words are more often real words of its language (word lists: wordfreq). A caller can switch between English and Hindi/Hinglish mid-call. Per-sentence choice on the held-out test split (first two caller sentences of each call, Piper TTS; `bench/results/langpick_test.json`):

Channel	English chosen for English speech	Hindi chosen for Hindi speech
telephone band, SNR 10–30 dB	93.0 % (80/86)	91.0 % (91/100)
near speaker	87.2 % (75/86)	99.0 % (99/100)

Overall 92.7 % (345/372). Most errors are short English sentences whose Hindi transcript is a 1–2 word fragment; borderline sentences keep the previous sentence’s language, and warnings follow the majority of the last three sentences. Not yet measured: how much a wrong choice changes the final warning.

On-phone benchmark (§4, `docs/benchmarks.md`), measured on the team’s phone on battery, 12 Nov 2026 (`bench/results/device/kavach_bench_20261112_013528.json`). Phone: Xiaomi 2312DRA50I, Snapdragon 7s Gen 2 (SM7435: 4 × Cortex-A78 2.4 GHz + 4 × Cortex-A55 1.96 GHz), 7.3 GB RAM, Android 15.

What	Result
Detector (rules + classifier + kill-chain)	4.07 ms per sentence
Hindi IndicConformer (sentence)	RTF 0.153 at 2 threads, 0.092 at 4 (6.5× / 10.9× faster than real time)
English Parakeet 110M (sentence)	RTF 0.052 at 2 threads, 0.037 at 4 (19× / 27×)
Small streaming models (fallback)	Hindi RTF 0.239, English 0.117 at 2 threads
Auto listening, one minute (VAD + both models on every sentence, 2 threads each)	**54 % of one core, 385 mA ≈ 7.6 % battery per hour**, battery 38.0 → 37.9 °C, thermal status 0, 235 MB
Llama 3.2 1B INT4 (ExecuTorch, XNNPACK)	load 4.1 s, +1.1 GB; first token 12.9 s cold (768-token prompt), **5.9 s with the warm system prefix**; prefill 56–60 tok/s; **decode 11.6–11.9 tok/s**; 858 mA while generating; no thermal throttling

Reading: speech recognition of both languages together uses about a fifth of real time, so the transcript keeps up with the caller; listening for a whole hour costs under 8 % battery. Llama is the slowest part, which is why its explanation always follows the instant rule-based warning. Four threads make the big models

clearly faster (Hindi 6.5× → 10.9×); the app uses two to save battery, a setting to revisit with a 4-thread battery measurement.

7.6 Network Watch

Part	Test	Result
App traffic watch	200 simulated 2-hour traces per scenario at 30 s polling	spyware photo / SMS uploads, slow exfiltration, heartbeats every 60 s, 5 min and jittered 2 min: **100 % detected** (median 30–90 s for uploads, 10–35 min for heartbeats); chat, video sharing, streaming, Play-store backup: **0 % false alarms**; a sideloaded backup app or an exact-timer sideloaded sync is always flagged (by design; “I know this app”)
Domain model	6 DGA families never seen in training	cryptolocker 92 %, necurs 92 %; dictionary-word DGAs (suppobox, matsnu, gozi, pushdo) ≈ 0 % (blind spot); 0.36 % false alarms on 85k popular domains, 0 genuine Indian domains flagged. A single name never alarms: ≥ 3 non-existent machine-generated names from one app in 10 min are required
Flow model, UNSW-NB15 test (82,332 flows)	phone-observable features, threshold for 2 % validation false alarms	P 0.939, R 0.885, **F1 0.911**, 7.1 % false alarms (all-features reference F1 0.934); Fuzzers only 41 %; leave-one-attack-out: Exploits 72 %, Reconnaissance 63 %, Fuzzers 30 %
Flow model, NSL-KDD	same features idea, KDDTest+ used once	F1 0.854; R2L 22 %, U2R 42 % (these need packet contents); new attack types 65 % vs seen 82 %
Flow model on phone-like traffic	replayed capture	flags **60/60 normal HTTPS sessions**: UNSW-NB15 has **zero normal TLS flows**. Kept as research, raises no alerts
Pcap replay demo	Scapy capture: browsing + scan + DGA + look-alike + heartbeat + 2.4 MB upload	rules find **all 5**; no alert on normal browsing
Home Wi-Fi check (Network Watch app)	TCP connect to 10 abused services (ADB 5555, Telnet, SMB, RDP, TR-069, FTP, RTSP, SSH, HTTP) on the /24	Android 14 emulator network: 254 addresses in 12 s; found the emulator’s open ADB port (urgent) with the fix in plain language

7.7 PS3 scope coverage

PS3 expectation	KAVACH component	Status at Phase 1
Scam / fraud call detection on device	Guardian (ASR → cues → classifier → kill-chain → Llama)	built; evaluated on synthetic, hand-written, held-out and noisy-audio sets
Phishing SMS / messages	Message Shield (notifications, SMS model, cues)	built, F1 0.939
Malicious links	Link Check (allow-list, look-alike rules, URL model)	built; 98.5 % of Indian look-alikes, 0 % of genuine Indian domains
Malicious / fake apps	App Inspector (permissions, sideloading, lure names)	v0 preview
Screen-sharing / remote-access fraud	Screen-share Guard + kill-chain	v0 preview; STOP on a payment app during remote access
Network threats / intrusion	Network Watch: app traffic, DNS watch, home Wi-Fi check, flow model	traffic and DNS rules built and evaluated; Wi-Fi check built; flow model research only
Cross-channel correlation	kill-chain hub (calls, messages, links, apps, network)	built, parity-tested
On-device, privacy-preserving	no INTERNET in the main apps; audio never stored	by construction
Explainable alerts in Indian languages	Hindi / English reasons, spoken warnings, Llama explanation	verified on the phone; Llama quality measured before/after LoRA (§7.9)
Arm optimisation	ExecuTorch + XNNPACK / KleidiAI, int8 ASR, big.LITTLE placement, < 1 ms fast path	export verified; on-phone numbers in Phase 2 week 1

7.8 Still to measure (scripts ready)

Measured on the team's phone (§7.5): speech speed at 1/2/4 threads, Llama load, first token, decode speed and memory, battery and temperature for an hour-equivalent of listening, and Llama quality before/after LoRA (§7.9). Still open: Llama per quantisation rung with KleidiAI on and off, on big and LITTLE cores and on a budget phone; Hindi WER on our own speaker-phone recordings; the 30-minute thermal soak; Llama verdict quality on the locked held-out calls; and the human-recorded test set (10–20 consented calls planned for 21 Nov). Reference point: ExecuTorch reports about 50 tok/s decode for Llama 3.2 1B SpinQuant on a OnePlus 12.

8. Limitations and threat model (summary)

- **ASR is the dominant error source.** Speaker-phone Hindi WER is unknown until recorded; a TV or other people talking near the phones is the hardest case (§7.3). Mitigations: the bigger English model, the placement check, victim-side cues, a Devanagari lexicon, training on ASR-noisy text, Llama over noisy text, and adaptation on our recordings (Phase 2).
- **Network Watch** sees byte counts only (the DNS app sees names, not contents); its rule numbers come from simulated traces, and the flow model needs phone traffic to be usable.
- **Synthetic training data**, written by the team. The held-out and hand-written sets reduce but do not remove author bias; the human-recorded test set is the real measure.
- **Adversaries** can avoid trigger words. The classifier, the cue-category combinations and Llama generalise beyond exact phrases, and cross-channel fusion catches the action (link, APK, screen share) even when the talk is clean.

- **Platform policy**: notification access, usage access and `QUERY_ALL_PACKAGES` suit a side-loaded prototype; `docs/threat_model.md` describes a Play-policy-compliant design.
- **The call must be on speaker** unless the phone maker enables system mode. KAVACH Phone puts unknown callers on speaker by itself. One-phone listening relies on the accessibility capture rule; Play policy limits accessibility use, so a store version would need the phone maker's system mode or a declared accessibility use case (`docs/threat_model.md`).
- **Hindi speech recognition** was the weakest link (§7.5). AI4Bharat IndicConformer closes most of the gap on synthetic audio; human Hindi voices on real phone calls are still to be measured.

7.9 Llama 3.2 1B after LoRA fine-tuning (Colab, 12 Nov 2026)

`notebooks/kavach_llama_lora_colab.ipynb` on a free Colab T4: LoRA r = 16 on all projections, 1 epoch over 4,000 KAVACH instruction pairs (4.1 h; final training loss 0.111, dev loss 0.041), merged and exported to ExecuTorch 1.4 with the shipped 4-bit config. Scored with `models/llama/eval_llama.py` on the first 100 dev examples, before and after (`bench/results/llama_lora_colab.json`):

Metric	Base Llama 3.2 1B	+ KAVACH LoRA
valid JSON verdict	71 %	**100 %**
scam / not scam correct	38 %	**89 %**
scam type correct	58 %	**76 %**
warning stage correct	71 %	**100 %**
quoted evidence really in the transcript	8 %	**96 %**
advice copied from the prompt's example	32 %	**0 %**
advice in the call's language	55 %	**100 %**

Caveats: the dev examples come from the same synthetic generator as the training data (the held-out test split was not used), and this is the full-precision merged model; the 4-bit `.pte` on the phone is checked on the phone (Calls → Setup & tools → Test on-device AI). The two problems seen on the phone earlier (copied example advice, English advice in a Hindi call) are exactly the ones that go to 0 % and 100 %.

On the phone (12 Nov, 04:24), the fine-tuned 4-bit `.pte` (820 MB, vs 1,083 MB for Meta's SpinQuant file) loaded and ran the self-test call: verdict "digital_arrest", confidence 0.91, its own advice ("Hang up. Do not transfer any money. Tell a family member. Call 1930 or report at cybercrime.gov.in.") instead of the prompt's example; 92 tokens, first token 18.3 s (cold, no warm prefix), 11.5 tokens/s decode. During a replayed Hindi digital-arrest call (05:10) it wrote its own Hindi advice ("यह डिजिटल अरेस्ट स्कैम है। पैसे देना बंद कीजिए। कोई 'पैसे ट्रांसफर' मत कीजिए। ... 1930 पर"), with verbatim evidence "साइबर क्राइम", "इंस्पेक्टर"; the screen labels it "Explained by Llama 3.2 1B on this phone". The Hindi is understandable but not always polished, and one earlier run on the same file was still replaced by the playbook text (the app now logs Llama's wording whenever that happens).

9. Phase 2 plan

1. Fine-tune the LoRA and export the quantisation ladder. A ready Colab notebook (`notebooks/kavach_llama_lora_colab.ipynb`, free T4, ≈2 h) runs data → base score → LoRA → trained score → ExecuTorch 1.4 `.pte`; `eval_llama.py` now also reports how often the advice copies the prompt's worked example and whether it is in the call's language. Measure quality, on-device speed and the **Llama escalation effect** on all call sets.

2. Record 100+ scam and 100+ genuine role-play calls (consented), measure Hindi WER, adapt ASR, and build the untouched human-recorded test set.
3. NPU export of the speech models (ExecuTorch QNN/MediaTek) and a CPU vs NPU energy comparison.
4. Gradient-boosted App Inspector on public malware data; retrain the Network Watch flow model on Android traffic (CIC-AndMal2017 or our own labelled captures); dictionary-DGA detection.
5. Budget-phone port, 30-minute thermal soak, and a consent-based pilot with 20–30 family volunteers.

10. Reproducing everything

```
        pip install numpy scipy scikit-learn pytest           # core + tests
        ./data/public/download.sh                             # public datasets
        python data/synth/generate_calls.py && python data/synth/generate_sms.py && python
data/dev/build_handwritten.py
        python models/classifiers/train_text.py && python models/classifiers/train_url.py
        python bench/eval_calls.py --split test               # + dev / handwritten / heldout
        python -m pytest -q tests && python tests/make_parity_vectors.py
        cd android && ./gradlew :core:test                     # Kotlin parity (over 330 checks)
        python models/netwatch/train_domain.py && python models/netwatch/train_ids.py
        python bench/netwatch_sim.py && python bench/netwatch_pcap_demo.py
        python bench/e2e_noise.py --asr zipformer-en-2023-06-21 --configs base hotwords
        scripts/build_apks.sh                                 # debug APKs (needs Android SDK)
        pip install executorch==1.4.0 transformers && python models/llama/tiny_pipeline_check.py
        pip install sherpa-onnx && python bench/e2e_audio.py --models <asr+tts model dir>
```

Sources: proposal §Sources (MHA/I4C figures, ExecuTorch Llama README, sherpa-onnx, RBI circular RBI/2025-26/28 on `.bank.in`), the model cards in `docs/model_cards.md` and the data card in `docs/data_card.md`.